Data_Visualization_Matplotlib

CODE

```
#When to use Which Library:

'''
■ Use Matplotlib when:
You want complete customization
■ Use Seaborn when:
You want quick, attractive statistical plots
■ Use Plotly when:
You need interactive dashboards or presentations
■ Use Bokeh when:
You need real-time or web-integrated visualization
■ Use ggplot when:
You prefer a structured, layered plotting style

'''
```

MARKDOWN

Let us Begin with Matplotlib

CODE

```
'''
What is pyplot?

pyplot is a module in Matplotlib that provides a simple, MATLAB-like interface for creating plots.

It allows you to:
Create figures
Plot data
Customize charts
Display visualizations '''
```

CODE

```
import matplotlib.pyplot as plt
```

CODE

```
#Understanding the Basic Plotting Workflow:
'''
Every plot in pyplot follows 3 simple steps:

Step 1: Create a Figure
Step 2: Plot the Data
Step 3: Show the Plot
'''
```

CODE

```
# Sample Data
x = [1, 2, 3, 4, 5]
y = [10, 20, 25, 30, 40]
```

CODE

```
#Create figure:
plt.figure()

'''
Creates a blank canvas
Think of it as a "drawing board"
'''
```

CODE

```
plt.figure(figsize=(6,4))
```

CODE

```
plt.plot(x, y)
plt.show()
```

CODE

```
'''What it does:
Draws a line graph
x → horizontal axis
y → vertical axis
'''
```

CODE

```
#Complete
import matplotlib.pyplot as plt

# Data
x = [1, 2, 3, 4, 5]
y = [10, 20, 25, 30, 40]

# Step 1: Create Figure
plt.figure(figsize=(6,4))

# Step 2: Plot Data
plt.plot(x, y)

# Step 3: Show Plot
plt.show()
```

CODE

```
plt.figure(figsize=(3,3))
plt.plot(x, y)

# Add labels
plt.xlabel("Days") # X-axis label
```

```
plt.ylabel("Revenue ($)") # Y-axis label

plt.show()
```

CODE

```
#Adding Title
plt.figure(figsize=(6,4))
plt.plot(x, y)

plt.xlabel("Days")
plt.ylabel("Revenue ($)")
plt.title("Daily Revenue Trend") # Plot title

plt.show()
```

CODE

```
# Plot with custom style
plt.plot(x, y, color='green', linestyle='--', marker='o', linewidth=2, markersize=8)

plt.xlabel("Days")
plt.ylabel("Revenue ($)")
plt.title("Daily Revenue Trend with Styles")

plt.show()
```

CODE

```
y2 = [8, 18, 22, 28, 35]

plt.figure(figsize=(6,4))

# Line 1
plt.plot(x, y, color='blue', linestyle='-', marker='o', label="Product A")
# Line 2
plt.plot(x, y2, color='red', linestyle='--', marker='s', label="Product B")

plt.xlabel("Days")
plt.ylabel("Revenue ($)")
plt.title("Revenue Trend for Multiple Products")
plt.legend() # Show legend

plt.show()
```

CODE

```
y2 = [8, 18, 22, 28, 35]

plt.figure(figsize=(6,4))

# Line 1
plt.plot(x, y, color='blue', linestyle='-', marker='o', label="Product A")
# Line 2
plt.plot(x, y2, color='red', linestyle='--', marker='s', label="Product B")
```

```
plt.xlabel("Days")
plt.ylabel("Revenue ($)")
plt.title("Revenue Trend for Multiple Products")
plt.legend() # Show legend

#Add a Grid
plt.grid(True)
plt.show()
```

CODE

```
plt.figure(figsize=(6,4))
plt.plot(x, y)

# Custom ticks
plt.xticks([1, 2, 3, 4, 5], rotation=90)
plt.yticks([0, 10, 20, 30, 40, 50])

plt.xlabel("Days")
plt.ylabel("Revenue ($)")
plt.title("Custom Tick Positions")

plt.show()
```

CODE

```
#Bar Plot (Vertical & Horizontal)
```

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv("retail_sales.csv")

# Quick look at data
df.head()
```

CODE

```
# Convert Date to datetime
df['Date'] = pd.to_datetime(df['Date'])

# Aggregate revenue by Date
daily_revenue = df.groupby('Date')['Revenue'].sum()
daily_revenue
```

CODE

```
#1. Line Plot
# Convert Date to datetime
df['Date'] = pd.to_datetime(df['Date'])
```

```
# Aggregate revenue by Date
daily_revenue = df.groupby('Date')['Revenue'].sum()

plt.figure(figsize=(10,5))
plt.plot(daily_revenue.index, daily_revenue.values, color='blue', marker='o', linestyle='-',
linewidth=2)

plt.title("Daily Total Revenue")
plt.xlabel("Date")
plt.ylabel("Revenue ($)")
plt.grid(True)
plt.xticks(rotation=45) # Rotate date labels
plt.ylim(0, max(daily_revenue.values)*1.2)
plt.show()
```

CODE

```
category_units = df.groupby('Product_Category')['Units_Sold'].sum()

# Vertical Bar
plt.figure(figsize=(6,4))
plt.bar(category_units.index, category_units.values, color=['skyblue','lightgreen','orange'])
plt.title("Total Units Sold by Category")
plt.xlabel("Product Category")
plt.ylabel("Units Sold")
plt.grid(axis='y')
plt.show()

# Horizontal Bar
plt.figure(figsize=(6,4))
plt.barh(category_units.index, category_units.values, color=['skyblue','lightgreen','orange'])
plt.title("Total Units Sold by Category (Horizontal)")
plt.xlabel("Units Sold")
plt.ylabel("Product Category")
plt.grid(axis='x')
plt.show()
```

CODE

```
plt.figure(figsize=(6,4))
plt.scatter(df['Units_Sold'], df['Profit'], color='purple', marker='^', s=50)
plt.title("Units Sold vs Profit")
plt.xlabel("Units Sold")
plt.ylabel("Profit ($)")
plt.grid(True)
plt.show()
```

CODE

```
region_revenue = df.groupby('Region')['Revenue'].sum()
colors = ['gold', 'lightblue', 'lightgreen', 'pink']

plt.figure(figsize=(6,6))
plt.pie(region_revenue.values, labels=region_revenue.index, colors=colors, autopct='%1.1f%%',
startangle=60 ,shadow=False)
```

```
plt.title("Revenue Share by Region")
plt.savefig("Pie.png")
plt.show()
```

CODE

```
import pandas as pd
df=pd.read_csv('sales_dataset.csv')
```

CODE

```
view(df)
```